

Feature Selection and Feature Extraction in Pattern Analysis: A Literature Review

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Abstract—Pattern analysis often requires a pre-processing stage for extracting or selecting features in order to help the classification, prediction, or clustering stage discriminate or represent the data in a better way. The reason for this requirement is that the raw data are complex and difficult to process without extracting or selecting appropriate features beforehand. This paper reviews theory and motivation of different common methods of feature selection and extraction and introduces some of their applications. Some numerical implementations are also shown for these methods. Finally, the methods in feature selection and extraction are compared.

Index Terms—Pre-processing, feature selection, feature extraction, dimensionality reduction, manifold learning.

I. INTRODUCTION

PATTERN recognition has made significant progress recently and is used for various real-world applications, from speech and image recognition to marketing and advertisement. Pattern analysis can be divided broadly into classification, regression (or prediction), and clustering, each of which tries to identify patterns in data in a specific way. The module which finds the pattern in data is named “model” in this paper. Feeding the raw data to the model, however, might not result in a satisfactory performance because the model faces a hard responsibility to find the useful information in the data. Therefore, another module is required in pattern analysis as a pre-processing stage to extract the useful information concealed in the input data. This useful information can be in terms of either better representation of data or better discrimination of classes in order to help the model perform better. There are two main categories of methods for this goal, i.e., feature selection and feature extraction.

To formally introduce feature selection and extraction, first, some notations should be defined. Suppose that there exist n data samples (points), denoted by $\{\mathbf{x}_i\}_{i=1}^n$, from which we want to select or extract features. Each of these data samples, $\mathbf{x}_i$, is a d -dimensional column vector (i.e., $\mathbf{x}_i \in \mathbb{R}^d$). The data samples $\{\mathbf{x}_i\}_{i=1}^n$ can be represented by a matrix $\mathbf{X} := [\mathbf{x}_1, \dots, \mathbf{x}_n] = [\mathbf{x}^1, \dots, \mathbf{x}^n]^\top \in \mathbb{R}^{d \times n}$. In supervised cases, the target labels are denoted by $\mathbf{t} := [t_1, \dots, t_n]^\top \in \mathbb{R}^n$. Throughout this paper, x , $\mathbf{x}$, $\mathbf{X}$, $\mathcal{X}$, and

$\mathcal{S}$ denote a scalar, column vector, matrix, random variable, and set, respectively. The subscript (x_i) and superscript (x^j) on the vector represent the i -th sample and the j -th feature, respectively. Therefore, x_i^j denotes the j -th feature of the i -th sample. Moreover, in this paper, $\mathbf{1}$ is a vector with entries of one, $\mathbf{1} = [1, 1, \dots, 1]^\top$, and $\mathbf{I}$ is the identity matrix. The ℓ_p -norm is also denoted by $\|\cdot\|_p$.

Both feature selection and extraction reduce the dimensionality of data [1]. The goal of both feature selection and extraction is mapping $\mathbf{x} \mapsto \mathbf{y}$ where $\mathbf{x} \in \mathbb{R}^d$, $\mathbf{y} \in \mathbb{R}^p$, and $p \leq d$. In other words, the goal is to have a better representation of data with dimension p , i.e., $\mathbf{Y} := [\mathbf{y}_1, \dots, \mathbf{y}_n] = [\mathbf{y}^1, \dots, \mathbf{y}^n]^\top \in \mathbb{R}^{p \times n}$. In feature selection, $p \leq d$ and $\{\mathbf{y}^j\}_{j=1}^p \subseteq \{\mathbf{x}^j\}_{j=1}^d$ meaning that the set of selected features is an inclusive subset of the original features. However, feature extraction tries to extract a completely new set of features (dimensions) from the pattern of data rather than selecting features out of the existing attributes [2]. In other words, in feature extraction, the feature set of $\{\mathbf{y}^j\}_{j=1}^p$ is not a subset of features of $\{\mathbf{x}^j\}_{j=1}^d$ but is a different space. In feature extraction, often $p \ll d$ holds.

II. FEATURE SELECTION

Feature selection [3], [4], [5] is a method of feature reduction which maps $\mathbf{x} \in \mathbb{R}^d \mapsto \mathbf{y} \in \mathbb{R}^p$ where $p < d$. The reduction criterion usually either improves or maintains the accuracy or simplifies the model complexity. When there are d number of features, the total number of possible subsets are 2^d . It is infeasible to enumerate the exponential number of subsets if d is large. Therefore, we need to come up with some method that works in a reasonable time. The evaluation of the subsets is based on some criterion, which can be categorized as filter or wrapper methods [3], [6], explained in the following.

A. Filter Methods

Consider a ranking mechanism used to grade the features (variables) and the features are then removed by setting a threshold [7]. These ranking methods are categorized as filter methods because they filter the features before feeding to a learning model. Filter methods are based on two concepts “relevance” and “redundancy”, where the former is

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dependence (correlation) of feature with target and the latter addresses whether the features share redundant information. In the following, the common filter methods are introduced.

1) **Correlation Criteria:** Correlation Criteria (CC), also known as Dependence Measure (DM), is based on the relevance (predictive power) of each feature. The predictive power is computed by finding the correlation between the independent feature x^j and the target (label) vector t . The feature with the highest correlation value will have the highest predictive power and hence will be most useful. The features are then ranked according to some correlation-based heuristic evaluation function [8]. One of the widely used criteria for this type of measure is Pearson Correlation Coefficient (PCC) [7], [9] defined as:

$$\rho_{x^j, t} := \frac{\text{cov}(x^j, t)}{\sqrt{\text{var}(x^j) \text{var}(t)}}, \quad (1)$$

where $\text{cov}(\cdot, \cdot)$ and $\text{var}(\cdot)$ denote the covariance and variance, respectively.

2) **Mutual Information:** Mutual Information (MI), also known as Information Gain (IG), is the measure of dependence or shared information between two random variables [10]. It is also described as Information Theoretic Ranking Criteria (ITRC) [7], [9], [11]. The MI can be described using the concept given by Shannon's definition of entropy:

$$H(X) := - \sum_x p(x) \log(p(x)), \quad (2)$$

which gives the uncertainty in the random variable X . In feature selection, we need to maximize the mutual information (i.e., relevance) between the feature and the target variable. The mutual information (MI), which is the relative entropy between the joint distribution and product distribution, is:

$$\text{MI}(X; T) := \sum_x \sum_t p(x^j, t) \log \frac{p(x^j, t)}{p(x^j)p(t)}, \quad (3)$$

where $p(x^j, t)$ is the joint probability density function of feature x^j and target t . The $p(x^j)$ and $p(t)$ are the marginal density functions. The MI is zero or greater than zero if X and Y are independent or dependent, respectively. For maximizing this MI, a greedy step-wise selection algorithm is adopted. In other words, there is a subset of features, denoted by matrix S , initialized with one feature and features are added to this subset one by one. Suppose Y_S denotes the matrix of data having features whose indices exist in S . The index of selected feature is determined as [12]:

$$j := \arg \max_{j \notin S} \text{MI}(Y_S \cup x^j; t). \quad (4)$$

The above equation is also used for selecting the initial feature. Note that it is assumed the selected features are independent. The stop criterion for adding the new features is when there is highest increase in the MI at the previous step. It is noteworthy that if a variable is redundant or not informative in isolation, this does not mean that it cannot be useful when combined with another variable [13]. Moreover, this approach can reduce the dimensionality of features without having any significant negative impact on the performance [14].

3) **χ^2 Statistics:** The χ^2 Statistics method measures the dependence (relevance) of feature occurrence on the target value [14] and is based on the χ^2 probability distribution [15], defined as $Q = \sum_{i=1}^k Z_i^2$, where Z_i 's are independent random variables with standard normal distribution and k is the degree of freedom. This method is only applicable to cases where the target and features can take only discrete finite values. Suppose $n_{p,q}$ denotes the number of samples which have the p -th value of the j -th feature and the q -th value of the target t . Note that if the j -th feature and the target can have p and q possible values, respectively, then $p = \{1, \dots, p\}$ and $q = \{1, \dots, q\}$. A contingency table is formed for each feature using the $n_{p,q}$ for different values of target and the feature. The χ^2 measure for the j -th feature is obtained by:

$$\chi^2(x^j, t) := \sum_{p=1}^p \sum_{q=1}^q \frac{(n_{p,q} - \mathbb{E}_{p,q})^2}{\mathbb{E}_{p,q}}, \quad (5)$$

where $\mathbb{E}_{p,q}$ is the expected value for $n_{p,q}$ and is obtained as:

$$\mathbb{E}_{p,q} = n \times \Pr(p) \times \Pr(q) = n \times \frac{\sum_{q=1}^q n_{p,q}}{n} \times \frac{\sum_{p=1}^p n_{p,q}}{n}. \quad (6)$$

The χ^2 measure is calculated for all the features. The large χ^2 shows the significant dependence of the feature to the target; therefore, if it is less than a pre-defined threshold, the feature can be discarded. Note that χ^2 Statistics face a problem when some of the values of features have very low frequency. The reason for this problem is that the expected value in equation (6) becomes inaccurate at low frequencies.

4) **Markov Blanket:** Feature selection based on Markov Blanket (MB) [16] is a category of methods based on relevance. MB considers every feature x^j and the target t as a node in a faithful Bayesian network. The MB of a node is defined as the parents, children, and spouses of that node; therefore, in a graphical model perspective, the nodes in MB of a node suffice for estimating that node. In MB methods, the MB of a target node is found, the features in that MB are selected, and the rest are removed. Different methods have been proposed for finding the MB of target, such as Incremental Association MB (IAMB) [17], Grow-Shrink (GS) [18], Koller-Sahami (KS) [19], and Max-Min MB (MMMB) [20]. The MMMB as one of the best methods in MB is introduced here. In MMMB, first, the parents and children of target node are found. For that, those features are selected (denoted by S) that have the maximum dependence with the target, given the subset of S with the minimum dependence with the target. This set of features includes some false positive nodes which are not parent or child of the target. Thus, the false positive features are filtered out if the selected features and the target are independent given any subset of selected features. Note that the G^2 test, Fisher correlation, Spearman correlation, and Pearson correlation can be used for the conditional independence test. Next, spouses of target are found to form the MB with the S (previously found parents and children). For this, the same procedure is done for the nodes in set S to find the spouses,

grandchildren, grandparents, and siblings of the target node. Everything except the spouses should be filtered out; thus, if a feature is dependent on the target given any subset of nodes having their common child, it is retained and the rest are removed. The parents, children, and spouses of the target found form the MB.

5) *Consistency-based Filter*: Consistency-Based Filters (CBF) use a consistency measure which is based on both relevance and redundancy and is a selection criterion that aims to retain the discrimination power of the data defined by original features [21]. The InConsistency Rate (ICR) of all features is calculated via the following steps. First, inconsistent patterns are found; these are defined as patterns that are identical but are assigned to two or more different class labels, e.g., samples having patterns (0, 1) and (0, 1) in two features and with targets 0 and 1, respectively. Second, the inconsistency count for a pattern of a feature subset is calculated by taking the total number of times the inconsistent pattern occurs in the entire dataset minus the largest number of times it occurs with a certain target (label) among all targets. For example, assume c_1 , c_2 , and c_3 are the number of samples having targets 1, 2, and 3, respectively. If c_3 is the largest, the inconsistency count will be $c_1 + c_2$. Third, ICR of a feature subset is the sum of inconsistency counts over all patterns (in a feature subset, there can be multiple inconsistent patterns) divided by the total number of samples n . The feature subset S with $\text{ICR}(S) \leq \delta$ is considered to be consistent, where δ is a pre-defined threshold. This threshold is included to tolerate noisy data. For selecting a feature subset S in CBF methods, FocusM and Automatic-Branch-and-Bound (ABB) are exhaustive search methods that can be used. This yields the smallest subset of consistent features. Las Vegas Filter (LVF), SetCover, and Quick-Branch-and-Bound (QBB) are faster search methods and can be used for large datasets when exhaustive search is not feasible.

6) *Fast Correlation-based Filter*: Fast Correlation-Based Filter (FCBF) [22] uses an entropy-based measure called Symmetrical Uncertainty (SU) to find correlation, for both relevance and redundancy, as follows:

$$\text{SU}(X, Y) := 2 \left(\frac{\text{IG}(X; Y)}{H(X) + H(Y)} \right), \quad (7)$$

where $\text{IG}(X; Y) = H(X) - H(X|Y)$ is the information gain and $H(X)$ is entropy defined by equation (2). The value $\text{SU}(x^i, t)$ between a feature x^i and target t is calculated for each feature. A threshold value for SU is used to determine whether a feature is relevant or not. A subset of features S is decided by this threshold. To find out if a relevant feature x^i is redundant or not, $\text{SU}(x^i, x^j)$ value between two features x^i and x^j is calculated. For a feature x^i , all its redundant features are collected together (denoted by S_{P_i}) and divided in two subsets, $S_{P_i}^+$ and $S_{P_i}^-$, where $S_{P_i}^+ = \{x^j | x^j \in S_{P_i}, \text{SU}(x^j, t) > \text{SU}(x^i, t)\}$ and $S_{P_i}^- = \{x^j | x^j \in S_{P_i}, \text{SU}(x^j, t) \leq \text{SU}(x^i, t)\}$. All features in $S_{P_i}^+$ are processed before making a decision on x^i . If a feature is predominant (i.e., its $S_{P_i}^+$ is empty) then all features in $S_{P_i}^-$ are removed and x^i is retained. The feature with the

largest $\text{SU}(x^i, t)$ value is a predominant feature and used as a starting point to eliminate other features. FCBF uses the novel idea of predominant correlation to make feature selection faster and more efficient so that it can be easily used on high dimensional data. This method greatly reduces dimensionality and increases classification accuracy.

7) *Interact*: In feature selection, many algorithms apply correlation metrics to find which feature correlates most to the target. These algorithms single out features and do not consider the combined effect of two or more features with the target. In other words, some features might not have individual effect but alongside other features they give high correlation to the target and increase classification performance. Interact [23] is a fast filter algorithm that searches for interacting features. First, Symmetrical Uncertainty (SU), defined by equation (7), is used to evaluate the correlation of individual features with the target. This heuristic is used to rank features in descending order such that the most important feature is positioned at the beginning of the list S . Interact also uses Consistency Contribution (cc) or c-contribution metric which is defined as:

$$\text{cc}(x^i, X) := \text{ICR}(X \setminus x^i) - \text{ICR}(X), \quad (8)$$

where x^i is the feature for which cc is being calculated, and ICR stands for inconsistency rate defined in Section II-A.5. The “ $\setminus$ ” is the set-theoretic difference operator, i.e., $X \setminus x^i$ means X excluding the feature x^i . The cc is calculated for each feature from the end of the list S and if the cc of a feature is less than a pre-defined threshold, that feature is eliminated. This backward elimination makes sure that the cc is first calculated for the features having small correlation with the target. The appropriate pre-defined threshold is assigned based on cross validation. The Interact method has the advantage of being fast.

8) *Minimal-Redundancy-Maximal-Relevance*: The Minimal Redundancy Maximal Relevance (mRMR) [24], [25] is based on maximizing the relevance and minimizing redundancy of features. The relevance of features means that each of the selected features should have the largest relevance (correlation or mutual information [26]) with the target t for having better discrimination [11]. This dependence or relevance is formulated as the average mutual information between the selected features (set S) [25]:

$$d := \frac{1}{|S|} \sum_{x^i \in S} \text{MI}(x^i; t), \quad (9)$$

where MI is the mutual information defined by equation (3). The redundancy of features, on the other hand, should be minimized because having at least one of the redundant features suffices for a good performance. The redundancy of features x^i and x^j is formulated as [25]:

$$r := \frac{1}{|S|^2} \sum_{x^i, x^j \in S} \text{MI}(x^i; x^j). \quad (10)$$

By defining $\phi = d - r$, the goal of mRMR is to maximize ϕ [25]. Therefore, the features which maximize ϕ are added to the set S incrementally and one by one [25].

B. Wrapper Methods

As seen previously, filter methods select the optimal features to be passed to the learning model, i.e., classifier, regression, etc. Wrapper methods, on the other hand, integrate the model within the feature subset search. In this way, different subsets of features are found or generated and evaluated through the model. The fitness of a feature subset is evaluated by training and testing it on the model. Thus in this sense, the algorithm for the search of the best suboptimal subset of the feature set is essentially “wrapped” around the model. The search for the best subset of the feature set, however, is an NP-hard problem. Therefore, heuristic search methods are used to guide the search. These search methods can be divided in two categories: Sequential and Metaheuristic algorithms.

1) **Sequential Selection Methods:** Sequential feature selection algorithms access the features from the given feature space in a sequential manner. These algorithms are called sequential due to the iterative nature of the algorithms. There are two main categories of sequential selection methods: Sequential Forward Selection (SFS) and Sequential Backward Selection (SBS) [27]. Although both the methods switch between including and excluding features, they are based on two different algorithms according to the dominant direction of the search. In SFS, the set of selected features, denoted by $\mathcal{S}$, is initially empty. The features are added to $\mathcal{S}$ one by one if they improve the model performance the best. This procedure is repeated until the required number of features are selected. In SBS, on the other hand, the set $\mathcal{S}$ is initialized by the entire features and features are removed sequentially based on the performance [3]. The SFS and SBS ignore the dependence of features, i.e., some specific features might have better performance than the case where those features are used alongside some other features. Therefore, Sequential Floating Forward Selection (SFFS) and Sequential Floating Backward Selection (SFBS) [28] were proposed. In SFFS, after adding a feature as in SFS, every feature is tested for being excluded if the performance improves. Similar approach is performed in SFBS but with opposite direction.

2) **Metaheuristic Methods:** The metaheuristic algorithms are also referred to as evolutionary algorithms. These methods have low implementation complexity and can adapt to a variety of problems. They are also less prone to get stuck in a local optima as compared to sequential methods, where the objective function is the performance of the model. Many metaheuristic methods have been used for feature selection such as the binary dragonfly algorithm used in [29]. Another recent technique called whale optimization algorithm was used for feature selection and explored against other common metaheuristic methods in [30]. As examples of metaheuristic approaches for feature selection, the two broad metaheuristic methods, Particle Swarm Optimization (PSO) and Genetic Algorithms (GA), are explained here. PSO is inspired by the social behavior observed in some animals like the flocking of birds or formation of schools of fish and GA is inspired by natural selection and has been used widely in search and

optimization problems.

The original version of PSO [31] was developed for continuous optimization problems whose potential solutions are represented by particles. However, many problems are combinatorial optimization problems usually with binary decisions. Geometric PSO (GPSO) uses a geometric framework described in [32] to provide a version of PSO that can be generalized to any solution representation. GPSO considers the current position of particle, the global best position, and the local best position of the particle as the three parents of particle and creates the offspring (similar to GA approach) by applying a three-Parent Mask-Based Crossover (3PMBCX) operator on them. The 3PMBCX simply takes each element of crossover with specific probabilities from the three parents. The GPSO is used in [33] for the sake of feature selection. Binary PSO [34] can also be used for feature selection where binary particles represent selection of features.

In GA, the potential solutions are represented by chromosomes [35]. For feature selection, the genes in the chromosome correspond to features and can take values 1 or 0 for selection or not selection of feature, respectively. The generations of chromosomes improve by crossovers and mutations until convergence. The GA is used in different works such as [36], [37] for selecting features. A modified version of GA, named CHCGA [38], is used in [39] for feature selection. The CHCGA maintains diversity and avoids stagnation of the population by using a Half Uniform Crossover (HUX) operator which crosses over half of the non-matching genes at random. One of the problems of GA is poor initialization of chromosomes. A modified version of GA proposed in [40] for feature selection reduces the risk of poor initialization by introducing some excellent chromosomes as well as random chromosomes for initialization. In that work, the estimated regression coefficients $\hat{\beta}_1, \dots, \hat{\beta}_d$ and their estimated standard deviations $\sigma_{\hat{\beta}_1}, \dots, \sigma_{\hat{\beta}_d}$ are calculated using regression, then Student's t -test is applied for every coefficient $t_j \leftarrow (\hat{\beta}_j - 0) / \sigma_{\hat{\beta}_j}$. The binary excellent chromosome $\mathbf{s} = [s_1, \dots, s_d]^T$ is generated where $s_j = 1$ with probability $|t_j| / \sum_{i=1}^d |t_i|$.

III. FEATURE EXTRACTION

As features of data are not necessarily uncorrelated (matrix $\mathbf{X}$ is not full rank), they share some information and thus there usually exists dummy information in the data pattern. Hence, in mapping $\mathbf{x} \in \mathbb{R}^d \mapsto \mathbf{y} \in \mathbb{R}^p$ usually $p \ll d$ or at least $p < d$ holds, where p is named the intrinsic dimension of data [41]. This is also referred to as the manifold hypothesis [42] stating that the data points exist on a lower dimensional sub-manifold or subspace. This is the reason that feature extraction and dimensionality reduction are often used interchangeably in the literature. The main goal of dimensionality reduction is usually either better representation/discrimination of data [43] or easier visualization of data [44], [45]. It is noteworthy that the $\mathbb{R}^p$ space is referred to as feature space (i.e., *feature extraction*), embedded space (i.e., *embedding*), encoded space

(i.e., *encoding*), subspace (i.e., *subspace learning*), lower-dimensional space (i.e., *dimensionality reduction*) [46], sub-manifold (i.e., *manifold learning*) [47], or representation space (i.e., *representation learning*) [48] in the literature.

The dimensionality reduction methods can be divided into two main categories, i.e., supervised and unsupervised methods. Supervised methods take into account the labels and classes of data samples while the unsupervised methods are based on the variation and pattern of data. Another categorization of feature extraction is dividing methods into linear and non-linear. The former assumes that the data falls on a linear subspace or classes of data can be distinguished linearly, while the latter supposes that the pattern of data is more complex and exists on a non-linear sub-manifold. In the following, we explore the different methods of feature extraction. For the sake of brevity, we mention the basics and forgo some very detailed methods and improvements of these methods such as using Nyström Method for incorporating out-of-sample data [49].

A. Unsupervised Feature Extraction

Unsupervised methods in feature extraction do not use the labels/classes of data for feature extraction [50], [51]. In other words, they do not respect the discrimination of classes but they mostly concentrate on the variation and distribution of data.

1) **Principal Component Analysis:** Principal Component Analysis (PCA) [52], [53] was first proposed by [54]. As a linear unsupervised method, it tries to find the directions which represent the variation of data the best. The original coordinates do not necessarily represent the direction of variation. The aim of PCA is to find orthogonal directions which represent the data with the least error [55]. Therefore, PCA can be also considered as rotating the coordinate system [56].

The projection of data onto direction $\mathbf{u}$ is $\mathbf{u}^\top \mathbf{X}$. Taking $\mathbf{S}$ to be the $d \times d$ covariance matrix of data, the variance of this projection is $\mathbf{u}^\top \mathbf{S} \mathbf{u}$. PCA tries to maximize this variance to find the most variant orthonormal directions of data. Solving this optimization problem [55] yields to $\mathbf{S} \mathbf{u} = \lambda \mathbf{u}$ which is the eigenvalue problem [57] for the covariance matrix $\mathbf{S}$. Therefore, the desired directions (columns of matrix $\mathbf{U}$) are the eigenvectors of the covariance matrix of data. The eigenvectors, sorted by their eigenvalues in descending order, represent the largest to smallest variations of data and are named principal directions or axes. The features (rows) of the projected data $\mathbf{U}^\top \mathbf{X}$ are named principal components. Usually, the components with smallest eigenvalues are cut off to reduce the data. There are different methods for estimating the best number of components to keep (denoted by p), such as using Bayesian model selection [58], scree plot [59], and comparing the ratio $\lambda_i / \sum_{j=1}^d \lambda_j$ with a threshold [53] where λ_i denotes the eigenvalue related to the i -th principal component.

In this paper, out-of-sample data refers to the data sample which does not exist in the training set of samples from which the subspace was created. Assume $\mathbf{U} = [\mathbf{u}_1, \dots, \mathbf{u}_d] \in$

$\mathbb{R}^{p \times d}$. The projection of training data $\mathbf{X}$ and the out-of-sample data $\mathbf{x}$ are $\mathbf{Y} = \mathbf{U}^\top \mathbf{X} \in \mathbb{R}^{p \times n}$ and $\mathbf{y} = \mathbf{u}^\top \mathbf{x}$, respectively. Note that the projected data can also be reconstructed back but it will have distortion error because the samples were projected onto a subspace. The reconstruction of training and out-of-sample data are $\hat{\mathbf{X}} = \mathbf{U} \mathbf{Y}$ and $\hat{\mathbf{x}} = \mathbf{U} \mathbf{y}$, respectively.

2) **Dual Principal Component Analysis:** The explained PCA was based on eigenvalue decomposition; however, it can be done based on Singular-Value Decomposition (SVD) more easily [55]. Considering $\mathbf{X} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^\top$, the $\mathbf{U}$ is exactly the same as before and contains the eigenvectors (principal directions) of $\mathbf{X} \mathbf{X}^\top$ (the covariance matrix). The matrix $\mathbf{V}$ contains the eigenvectors of $\mathbf{X}^\top \mathbf{X}$. In cases where $d \gg n$, such as in images, calculating eigenvectors of $\mathbf{X} \mathbf{X}^\top$ with size $d \times d$ is less efficient than $\mathbf{X}^\top \mathbf{X}$ with size $n \times n$. Hence, in these cases, dual PCA is used rather than the ordinary (or direct) PCA [55]. In dual PCA, projection and reconstruction of training data ($\mathbf{X}$) and out-of-sample data ($\mathbf{x}$) are formulated as:

$$\text{Projection for } \mathbf{X} : \quad \mathbf{Y} = \mathbf{\Sigma} \mathbf{V}^\top, \quad (11)$$

$$\text{Projection for } \mathbf{x} : \quad \mathbf{y} = \mathbf{\Sigma}^{-1} \mathbf{V}^\top \mathbf{X}^\top \mathbf{x}, \quad (12)$$

$$\text{Reconstruction for } \mathbf{X} : \quad \hat{\mathbf{X}} = \mathbf{X} \mathbf{V} \mathbf{V}^\top, \quad (13)$$

$$\text{Reconstruction for } \mathbf{x} : \quad \hat{\mathbf{x}} = \mathbf{X} \mathbf{V} \mathbf{\Sigma}^{-2} \mathbf{V}^\top \mathbf{X}^\top \mathbf{x}. \quad (14)$$

3) **Kernel Principal Component Analysis:** PCA tries to find the linear subspace for representing the pattern of data. However, kernel PCA [60] finds the non-linear subspace of data which is useful if the data pattern is not linear. The kernel PCA uses kernel method which maps data to a higher dimensional space $\mathbf{x} \mapsto \phi(\mathbf{x})$ where $\phi : \mathbb{R}^d \rightarrow \mathbb{R}^{d'}$ and $d' > d$ [61], [62]. Note that having high dimensions has both its blessings and curses [63]. The “curse of dimensionality” refers to the fact that by going to higher dimensions, the number of samples required for learning a function grows exponentially. On the other hand, the “blessing of dimensionality” states that in higher dimensions, the representation or discrimination of data is easier. Kernel PCA relies on the blessing of dimensionality by using kernels.

The kernel matrix is $K(\mathbf{x}_1, \mathbf{x}_2) = \phi(\mathbf{x}_1)^\top \phi(\mathbf{x}_2)$ which replaces $\mathbf{x}_1^\top \mathbf{x}_2$ using the kernel trick. The most popular kernels are linear kernel $\mathbf{x}_1^\top \mathbf{x}_2$, polynomial kernel $(1 + \langle \mathbf{x}_1, \mathbf{x}_2 \rangle)^i$, and Gaussian kernel $\exp(-\|\mathbf{x}_1, \mathbf{x}_2\|_2^2 / 2\sigma^2)$, where i is a positive integer and denotes the polynomial grade of kernel. Note that Gaussian kernel is also referred to as Radial Basis Function (RBF) kernel.

After the kernel trick, PCA is applied on the data. Therefore, in kernel PCA, SVD is applied on the kernel matrix $K(\mathbf{x}_1, \mathbf{x}_2) = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^\top$ rather than on $\mathbf{X}$. The projections of training and out-of-sample data are formulated as:

$$\text{Projection for } \mathbf{X} : \quad \mathbf{Y} = \mathbf{\Sigma} \mathbf{V}^\top, \quad (15)$$

$$\text{Projection for } \mathbf{x} : \quad \mathbf{y} = \mathbf{\Sigma}^{-1} \mathbf{V}^\top K(\mathbf{X}, \mathbf{x}). \quad (16)$$

Note that reconstruction cannot be done in kernel PCA [55]. It is noteworthy that kernel PCA usually does not perform satisfactorily in practice [55], [64] and the reason of it is